

Coverage Was Never the Binding Constraint: Break-Even Arithmetic and a Degree-Indexed Screen for Cold-Start Retrieval

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Abstract

A recent multi-domain study measures a retrieval bottleneck in LLM-based cold-start recommendation: a single retriever places the gold item in a 200-item pool only 4.6–22.9% of the time, and eight coverage-side mitigations all fail or are modest. We ask what a pool slot is actually worth to a cold item. Under *forced coverage* (the gold item injected into every pool, so retriever selection bias cannot enter, and the residual bias runs against our conclusion), the pre-registered conversion ratio $\rho(k) = R_{\text{cold}}(k)/R_{\text{warm}}$ has its upper confidence bound below 0.5 in every training-degree bucket $k \leq 10$ on two datasets, with $\rho(0) \approx 0.0002$. Substituting $\rho(k)$ into the corrected break-even prevalence $\pi^* = c/(g\rho + c)$, with the published coverage trades as constants, gives $\pi^*(k \leq 10, K=200)$ of 77–100%: no plausible cold prevalence pays. We then pre-registered an end-to-end validation of the resulting admission rule on quota-mixed pools, and it **failed on both datasets**: measured deltas are positive at moderate mixing while the segment-additive prediction is negative. The gain never flows through the cold items, whose end-to-end success stays ≈ 0 at every mixing level; it comes from competitor-set displacement among warm targets. Our position is therefore two-sided. Conversion, not coverage, binds for cold items, and no mixing level rescues them. But coverage $\times$ conversion ledgers, naive or corrected, mispredict the *sign* of mix-level interventions, because conversion is pool-composition-dependent. $\pi^*(k, K)$ is a necessary-but-insufficient screen, never a deploy signal; conversion must be measured in situ.

CCS Concepts

• Information systems $\rightarrow$ Recommender systems; Evaluation of retrieval results.

Keywords

cold start, candidate retrieval, coverage, evaluation, pre-registration

1 Introduction

Cold-start evaluation recently acquired a measured bottleneck. Across five domains, a single retriever puts the gold item into a 200-item candidate pool only 4.6–22.9% of the time, and 32–91% of cold-start targets are never covered at all [7]. The same study prices the fix: 5 $\times$ upweighting of item-new positives buys +10pp new-item coverage on Yelp at –4.5pp warm coverage, and +2.2pp on Video Games at –4.7pp. It states, and does not attempt to answer, the decision question: at what cold-target prevalence does that trade pay?

The naive answer is one line. An intervention with cold gain g and warm cost c pays iff prevalence π exceeds $\pi^* = c/(g + c)$.

Coverage is a per-event average over the same cold/warm partition as NDCG, so the identity transfers exactly and the published trades give 31.0% (Yelp) and 68.1% (Video Games). This is wrong, and not in the algebra. NDCG is (approximately) the objective; coverage is an upper bound on it. Writing end-to-end success as coverage times the conditional conversion rate $R = P(\text{top-}k \mid \text{in pool})$, standard two-stage analysis that we do not claim, and setting the change in success to zero gives

$$\pi^* = \frac{c}{g\rho + c}, \quad \rho = \frac{R_{\text{cold}}}{R_{\text{warm}}}, \quad (1)$$

of which the naive form is the special case $\rho = 1$. Everything hinges on ρ . The most replicated finding of our own earlier measurements is that strictly cold items are unrankable: cold NDCG@10 is exactly 0.00000 across two loss families and five training arms, and semantic-ID generative retrieval independently reports exactly 0.00000 [5]. If $\rho \approx 0$ then $\pi^* \rightarrow 1$ and no prevalence justifies the trade.

But “strictly cold” is $k = 0$, not the item-new segments of published benchmarks, so the right object is the degree-indexed curve $\rho(k)$. Measuring it naively is biased in our own favour: a retriever covers the easy cold items first, so conversion conditional on *natural* coverage is computed on a favourable subset. We measure that inflation at roughly 2 $\times$ in mid buckets. The instrument therefore forces coverage, injecting the gold item into every pool, which removes retriever selection entirely. The residual bias is one-sided and runs against us: forced coverage averages over items no real retriever reaches, so it overestimates ρ and underestimates π^* . If even this optimistic ρ yields π^* near 1, the conclusion holds a fortiori.

All results were pre-registered with frozen falsification rules, and three frozen predictions failed: the 90% bar on π^* (partially); the end-to-end validation of the admission rule (on both datasets); and a drift prediction, in the instrument’s favour. We report all three unsoftened and execute the frozen consequence: the headline demotes to the measurement claims and the admission rule to a necessary-but-insufficient screen. Our contribution is the arithmetic, the degree-resolved measurement of its inputs, and the measured boundary of its validity.

2 The break-even rule

Original form. With cold gain $g > 0$ and warm cost $|c| > 0$ in NDCG@10 and cold prevalence π , the expected change $E = \pi g - (1 - \pi)|c|$ is an arithmetic identity, and $E = 0$ at $\pi^* = |c|/(g + |c|)$.

Naive port to coverage. Coverage is $P(\text{gold} \in \text{pool})$, also a per-event average over the same partition, so $\pi_{\text{cov}}^* = c_{\text{cov}}/(g_{\text{cov}} + c_{\text{cov}})$: 31.0% for Yelp and 68.1% for Video Games under the published trades [7]. The port is wrong because maximising coverage equals maximising end-to-end quality only if a covered cold item converts at the warm rate.

Table 1: π^* under the published coverage trades as a function of ρ .

ρ	Yelp π^*	Video Games π^*
1 (naive)	31.0%	68.1%
0.1	81.8%	95.5%
0.01	97.8%	99.5%
$\rightarrow 0$	$\rightarrow 100\%$	$\rightarrow 100\%$

Corrected form. Let $R_s = P(\text{top-}k \mid \text{gold} \in \text{pool})$ for segment s , so end-to-end success is $S_s = \text{Cov}_s \cdot R_s$. Holding R fixed (an assumption we bound in §3 and then falsify on the warm side in §4) and setting $\Delta S = 0$ gives Eq. 1. Its sensitivity to ρ is the whole point (Table 1). Even at $\rho = 1$, Video Games already needs 68% cold prevalence. This also gives a single mechanism for the eight failed mitigations in [7]: every one attacks Cov, none attacks R , and their own positive-controlled regime (coverage forced to ceiling) still finds LLM rerankers failing to beat CF/content baselines, a direct and low read on R . Two empirical questions remain: $\rho(k)$, degree by degree, and whether the segment-additive form survives a real intervention. §4 answers the first affirmatively and the second negatively.

3 The instrument

Target. $\rho(k) = R(k)/R_{\text{warm}}$ with $R = P(\text{gold} \in \text{top-10 of the pool} \mid \text{gold} \in \text{pool})$, sliced by the target item’s pre-cutoff training degree in buckets 0, 1–2, 3–5, 6–10, 11–20, 21–50, 51–200, >200.

Forced coverage. The gold item is injected into every pool; this is the positive-controlled regime of [8] used as a measurement instrument rather than a benchmark condition, and gold injection itself is the sampled-metrics tradition [11]. Natural-coverage conditioning is disqualifying, not merely untidy: it computes R on the subset the retriever already found easy, which rises artificially as coverage expands. Forced coverage removes that selection; its residual bias overestimates ρ .

Vacuity in single-stage scorers. A pool built from the ranker’s own top- $(K-1)$ plus gold is degenerate: pool-hit@10 reduces to full-catalogue hit@10. Coverage and conversion are separable only when retriever and ranker are different models, which is why [7], whose generators and rerankers are distinct, could see the split at all.

Design. Retriever $Q \neq$ ranker M . Pool = $\text{top}_{K-1}(Q) \cup \{\text{gold}\}$; M scores the pool; the endpoint is gold’s hit@10 within the pool. Two retrievers, both reported: Q_{ext} (frozen sentence-embedding title k NN, the cold-capable family) and Q_{pop} (global popularity, which structurally cannot reach cold items). $K \in \{100, 200, 500, 1000\}$, $K=200$ primary (the setting of [7]); π^* is always reported as $\pi^*(k, K)$. R_{warm} is the event-weighted pooled hit@10 over $k \geq 51$ within the same retriever’s pools; ρ is never transported across retrievers. The ranker is a sequential recommender (HSTU-style encoder with frozen text features, chunked full cross-entropy), evaluated only; no training. Datasets: Amazon Musical Instruments (MI; $n=5$ seeds) and Steam ($n=3$, raised to 5 in the validation stage), under a global-time protocol. Two structural gates must pass in every run: G1, pool-hit $\geq$ full-catalogue hit; G2, pool-hit

Table 2: T1: conversion ratio $\rho(k)$ under forced coverage (95% CI where $k \leq 10$).

bucket	MI $d=256$	MI $d=64$	Steam $d=256$
$k=0$	0.0002 [0.0000, 0.0004]	0.0003 [0.0000, 0.0007]	0.0001 [−0.0001, 0.0003]
$k1-2$	0.0154 [0.0132, 0.0176]	0.0093 [0.0053, 0.0134]	0.0328 [0.0142, 0.0515]
$k3-5$	0.0684 [0.0597, 0.0770]	0.0469 [0.0192, 0.0747]	0.0253 [−0.0139, 0.0644]
$k6-10$	0.1340 [0.1216, 0.1463]	0.1110 [0.0613, 0.1607]	0.0511 [0.0423, 0.0599]
$k11-20$	0.2437	0.2192	0.1036
$k21-50$	0.4636	0.4770	0.1559
$k51-200$	0.7729	0.7950	0.2411
$k>200$	1.4544	1.4101	1.0871

Table 3: T2: $\pi^*(k, 200)$ from $\rho(k)$ and the published trades.

bucket	MI×Yelp	MI×VG	Steam×Yelp	Steam×VG
$k=0$	100.0%	100.0%	100.0%	100.0%
$k1-2$	96.7%	99.3%	93.2%	98.5%
$k3-5$	86.8%	96.9%	94.7%	98.8%
$k6-10$	77.1%	94.1%	89.8%	97.7%
$k11-20$	64.9%	89.8%	81.3%	95.4%
$k21-50$	49.3%	82.2%	74.3%	93.2%

monotone in K . The falsification rule was frozen before any evaluation: the thesis is refuted if $\rho(k) > 0.5$ in any bucket at $k \leq 10$ on either dataset.

Amendment A1 (frozen before any validation run). Because the confirmatory π^* table transplants the trades of [7] onto our ρ , we added an own-dataset end-to-end validation: E1, own quota-mix coverage trades; E2 (primary), frozen prediction P1 that $\text{sign}(\Delta S_{\text{meas}}) = \text{sign}(\Delta S_{\text{pred}})$ at every mixing level λ above seed noise, with the frozen consequence that a P1 failure refutes deployability and demotes the headline; E3, Steam at $n=5$; E4, coverage-unit temporal drift.

4 Results

Text retriever, $K=200$, event-stride 2, two-sided t CIs at $\alpha=0.05$; both structural gates passed in all 13 confirmatory runs; the analyzer was frozen before any result was read.

C1 (primary, pre-registered): supported. Every $k \leq 10$ bucket has its upper CI bound below 0.5 on every set (Table 2). Free coverage buys a sub-threshold item at most 13% of a warm item’s conversion (MI $k6-10$) and under 4% at $k \leq 2$. ρ rises monotonically in k and head items exceed the pooled warm reference, as they must. E3 (Steam $n=5$) reproduced every $n=3$ bucket inside its CI ($k0/1-2/3-5/6-10 = 0.0001/0.0242/0.0236/0.0442$).

T2. Break-even prevalence $\pi^*(k, 200) = c/(g\rho(k) + c)$ with the cited trades (Yelp $g=+10\text{pp}$, $c=4.5\text{pp}$; Video Games $g=+2.2\text{pp}$, $c=4.7\text{pp}$) is given in Table 3. **C2 (pre-registered: $\pi^*(k \leq 10, 200) > 90\%$ under both trades): partially failed, reported as failed.** The bar holds under the VG trade (94.1–100%) but fails under the Yelp trade at MI $k3-5$ (86.8%), MI $k6-10$ (77.1%) and marginally at Steam $k6-10$ (89.8%). The substantive point survives, since 77–100% exceeds any plausible cold share, but the round-number bar was mis-set; it should have been derived from a stated prevalence ceiling. We do not restate it.

T3. Selection bias of natural coverage (C3, descriptive): confirmed. On MI $d=256$, $R_{\text{nat}}/R_{\text{forced}} = 0.2333/0.1063 = 2.2\times$

at k_{11-20} , $0.3947/0.2024 = 2.0\times$ at k_{21-50} , and $1.8\times$ at k_{51-200} . Conditioning on natural coverage, which is what an evaluation does when it measures conversion at each arm's own coverage, inflates apparent mid-bucket cold conversion by about $2\times$.

The screen $\hat{k}(K)$. At an illustrative $\pi = 40\%$ (inside the 38–78% cold-target prevalence range of our temporal windows), the smallest MI bucket passing under the Yelp trade is k_{51-200} ($\pi^* \approx 37\%$), so $\hat{k} \approx 51$; under the VG trade even k_{21-50} fails (82.2%). A bucket with $\pi < \pi^*(k, K)$ cannot pay for its slot through its own conversions. E2 shows that passing the screen licenses nothing further.

E1. Own-dataset trades. Deterministic across seeds (coverage depends only on the frozen retrievers). MI at $\lambda=0.2$: every sub- k_{50} bucket gains $+1.3-2.1$ pp coverage while k_{51-200} loses -5.36 pp; Steam is similar with the head paying -2.13 pp.

E2. Sign-agreement validation P1: refuted on both datasets. Quota-mixed pools (text priority, popularity fill, $K=200$, no forced gold), $\lambda \in \{0 \dots 1\}$, ΔS = change in end-to-end hit@10 (Table 4). Sign agreement is 0/5 seeds at every $\lambda \leq 0.5$ on both datasets, and 5/5 only at the pure-text endpoint $\lambda=1.0$; the secondary magnitude prediction also fails ($4-5\times$ over-pessimistic). Per the frozen consequence, forced-coverage R does not transfer to realistic mixed pools and $\pi^*(k, K)$ is not deployable as a slot-mix predictor.

What the failure reveals. Mixing text slots into a popularity pool *improves* end-to-end success at moderate λ even though every marginal covered cold item converts at $\rho \approx 0$; cold end-to-end success stays ≈ 0 at every λ . The gain comes from the swap changing the competitor set for still-covered warm targets. R is pool-composition-dependent (the same gold items convert at 0.29 under popularity pools versus 0.64 under text pools at the head), so segment-additive coverage $\times$ conversion accounting mispredicts even the sign; a competition-displacement term dominates at realistic mixing. This indicts the naive ledger and our corrected break-even equally.

E4. Coverage-unit drift: not confirmed, in the instrument's favour. $\pi^*(k)$ moves ≤ 1.7 pp between adjacent temporal windows on MI (predicted > 5 pp in half the buckets, by analogy with the $+19-31$ pp NDCG-unit drift we had measured earlier). The deployability failure is structural, not drift. $\rho(k)$ is also nearly loss-invariant (gBCE 0.0050 vs full-CE 0.0154 at k_{1-2}).

5 Related work and claims

The coverage bottleneck [7] is measured across five domains with eight mitigations (LLM scale 8B $\rightarrow$ 70B, dense retrievers, two-tower, rank fusion, static cold-aware allocation, graph-evidence prompts, tail-prior injection, LLM reranking), all failing or modest, and prices the regime conflict as a measured trade with, in its own words, no break-even analysis, Pareto frontier or constrained optimisation. Its conditional metrics are stratified by coarse coldness bucket, not training degree. We take its trade as cited constants and supply the decision formalism it does not attempt.

Coverage versus conversion. The decomposition is standard two-stage analysis; [8] already separates reranking quality from retrieval coverage with a positive-controlled regime, and gold injection is the sampled-metrics tradition analysed by [11].

We claim neither; we claim the degree-resolved curve $R(k)$, its warm-normalised ratio $\rho(k)$, and the chain into $\pi^*(k, K)$.

Allocation of retrieval budget. Learned, downstream-utility-aware allocation of candidate-generation budget across channels exists [6], as do online-LP exposure duals [1], explicit per-route quotas [9] and hand-set per-category caps [10]. These allocate on the channel or route axis by learned or tuned mechanisms; we derive a closed form on the cold/warm axis from cited constants plus one measured curve, and pair it with a pre-registered validation whose failure is a first-class result: the inputs are temporally stable in coverage units (E4) but not composition-transportable (E2).

Production admission gates exist and are tuned: a graduation filter after $\geq n$ consumptions with n set operationally [2], and exposure-threshold growth phases [3]. Ours is, to our knowledge, the first *measurement-derived* screen, and after E2 it is a screen, not a gate.

Thresholds and the unrankability floor. Interventional per-item degree variation places the cold-warm transition at 6–15 interactions and proposes no rule for acting on it [4]; we cite it as corroboration that the structure in k lies below ~ 16 . The floor that makes the screen bite (strictly cold items unrankable) is replicated in semantic-ID generative retrieval at exactly 0.00000 [5], i.e. $R_{\text{cold}} \approx 0$ in a third architecture family measured by authors with no stake in our thesis.

Claims. (1) The measurements: $\rho(k)$ under forced coverage with its pre-registered outcome and the $\sim 2\times$ natural-coverage selection bias. (2) The closed form $\pi^*(k, K)$ and the induced screen $\hat{k}(K)$, demoted per the frozen consequence from admission rule to necessary-but-insufficient screen. (3) The pre-registered refutation: segment-additive coverage $\times$ conversion ledgers, naive or corrected, mispredict the sign of quota-mix interventions because R is pool-composition-dependent. (4) A single mechanism for the eight failures in [7]: all attack coverage while cold conversion binds, with the caveat that mix interventions can still pay through warm-side displacement. **Non-claims:** the decomposition, the forced-coverage protocol, the allocation framing of [6], and any algorithmic fix. This is a measurement, a screen and a refutation, not a method.

6 Limitations

(1) R is not intervention-invariant: the instrument's bound covers only the cold segment's own R , and E2 shows warm R shifts with composition strongly enough to flip the sign of end-to-end effects. (2) The $k=0$ bucket is degenerate (prior work already reports exact zero there) and is flagged, not leaned on. (3) K is free; c_{cov} depends on pool size, so all headline numbers are $K=200$. (4) ρ is pool-composition-relative and must be re-measured in any target setup, never imported. (5) The trade constants are one paper's setup. (6) One ranker family; generative rankers untested here. (7) Two datasets, $n=3$ in the Steam confirmatory matrix (raised to 5 in E3), so one bucket's lower bound crosses zero in Table 2. (8) One intervention family: E2 mixes retrievers at evaluation time; coverage-aware *retraining*, the intervention of [7], changes which items are covered and how they are scored and could have

Table 4: E2: measured vs. predicted end-to-end ΔS under quota mixing ($\pm$ = seed SD).

λ	MI meas	MI pred	Steam meas	Steam pred
0.2	+0.00129 $\pm$ 0.00050	-0.00124 $\pm$ 0.00011	+0.00140 $\pm$ 0.00003	-0.00540 $\pm$ 0.00017
0.4	+0.00195 $\pm$ 0.00057	-0.00261 $\pm$ 0.00022	+0.00108 $\pm$ 0.00021	-0.01476 $\pm$ 0.00046
1.0	-0.00722 $\pm$ 0.00047	-0.03480 $\pm$ 0.00140	-0.01771 $\pm$ 0.00178	-0.07190 $\pm$ 0.00220

different composition effects. Corrections log: three frozen predictions failed (C2 partial, P1, E4) and are reported without post-hoc rescue.

7 Position

Conversion, not coverage, binds for cold items, and no mixing level rescues them. Under forced coverage $\rho(k \leq 10)$ never reaches half the warm rate, and in realistic quota-mixed pools cold end-to-end success stays ≈ 0 at every λ on both datasets. Coverage handed to cold items, free or bought, does not make them convert.

The ledger is wrong anyway. Both the naive $\pi^* = c/(g + c)$ and our corrected $\pi^* = c/(g\rho + c)$ mispredict the sign of mix-level interventions, because swapping pool slots changes the competitor set for still-covered warm targets, and that displacement term is first-order at moderate λ . A mix intervention can pay end-to-end while every marginal covered cold item converts at $\rho \approx 0$.

For practitioners: measure conversion in situ, under the pool composition you would deploy. The instrument here is the cheap first pass (eval-only, gold injection, two structural gates) and yields $\hat{k}(K)$ as a screen: an item failing it cannot pay for its slot through its own conversions; passing it is never a deploy signal.

For researchers: report coverage gains only with conversion-conditioning and composition controls. A coverage number is an upper bound, not an outcome; natural-coverage conversion flatters the mid buckets by $\sim 2\times$, and even correctly measured per-segment conversion does not add across segments once pool composition shifts.

References

- [1] Anon. 2023. Multi-channel Integrated Recommendation with Exposure Constraints. arXiv:2305.12319.
- [2] Anon. 2023. Multi-Funnel Fresh-Content Recommendation. In *KDD*. arXiv:2306.01720.
- [3] Anon. 2025. Exposure-Threshold Growth-Phase Gating for Cold-Start Items at Kuaishou. In *WWW*. DOI 10.1145/3701716.3715205.
- [4] Anon. 2025. Recommendation Is a Dish Better Served Warm. arXiv:2508.07856.
- [5] Anon. 2026. Can Generative Recommendation Reach Cold Items? A Temporal Perspective on Semantic-ID Generation. arXiv:2607.21101.
- [6] Anon. 2026. CAPTS: Channel-Aware, Preference-Aligned Trigger Selection for Multi-Channel Item-to-Item Retrieval. arXiv:2602.12564.
- [7] Anon. 2026. Diagnosing and Mitigating Retrieval Bottlenecks in LLM-Based Cold-Start Recommendation. arXiv:2606.29947.
- [8] Anon. 2026. Dual-Regime Evaluation of LLM Rerankers for Recommendation. arXiv:2604.16318.
- [9] Anon. 2026. Multi-Decoder OneRec. arXiv:2607.26500.
- [10] Anon. 2026. RealRoute: Adaptive Caps for Preference-Conditioned Retrieval Routing. arXiv:2604.20860.
- [11] Michael D. Ekstrand et al. 2023. Sampled Metrics and Their Pitfalls in Recommender Evaluation. arXiv:2309.11723.